

RDK Exclusive Command Usage

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1. hnut_boardid

The **hnut_boardid** command is used to retrieve the ID of the current development board (different development boards have different IDs).

⚠ The board ID affects hardware initialization during boot—please configure it with caution.

- **Example command output:**

```
root@ubuntu:~# hnut_boardid
0x6A84
```

- **Output explanation:**

0x	6A	8	4
	^^	^	^
Chip ID	Board Power Design	Board Design	Version

2. hnut_ps

The **hnut_ps** command prints process status information not supported by BusyBox's `ps` command, including process ID, parent process ID, priority, memory usage, virtual memory, and more.

- **Syntax**

```
hnut_ps
```

- **Supported Information**

- **pid**: Process ID. Each process in the operating system has a unique identifier called the process ID (pid), which is used to uniquely identify and reference a process within the system.
- **ppid**: Parent Process ID. This indicates the parent process that created the current process.
- **state**
 - : Process state.
 - **I**: Idle.
 - **R**: Running.
 - **S**: Sleeping.
 - **D**: Disk sleeping (uninterruptible sleep).
 - **T**: Stopped.
 - **X**: Dead.
 - **Z**: Zombie.
 - **t**: Tracing stop.
 - **P**: Parked.
- **prio**: Priority. Represents the scheduling priority of the process, usually expressed as a numeric value. A higher number typically indicates higher priority, making the process more likely to receive CPU time slices.
- **nice**: Nice value. Represents the scheduling priority as an integer. A lower nice value indicates higher priority, allowing the process to obtain CPU time more frequently.
- **rt_prio**: Real-time priority. Indicates the priority of real-time processes; a lower value signifies higher real-time priority.
- **policy**: Scheduling policy. Specifies the scheduling policy used for the process, typically a scheduling algorithm such as First-In-First-Out (FIFO), Round Robin (RR), etc.
- **vsz**: Virtual memory size. Represents the size of the virtual memory address space accessible to the process.
- **rss**: Resident Set Size (physical memory usage). Indicates the amount of physical RAM currently allocated and used by the process.
- **comm**: Command name. Contains the name of the command or executable file associated with the process, used to identify the process type or purpose.

3. hrut_somstatus

The **hrut_somstatus** command retrieves temperature sensor readings, CPU/BPU operating frequencies, and BPU load.

• Syntax

```
sudo hrut_somstatus
```

• Common Usage

```
sunrise@ubuntu:~$ sudo hrut_somstatus
=====1=====
temperature-->
    pvt_cmn_pvtcl_t1 : 38.565 (C)
    pvt_cmn_pvtcl_t2 : 40.219 (C)
    pvt_mcu_pvtcl_t1 : 37.829 (C)
```

```

pvt_mcu_pvtc1_t2 : 38.013 (C)
pvt_bpu_pvtc1_t1 : 38.932 (C)
voltage-->
FAKE      : 100.0 (mV)
VDDQ_DDR2 : 498.0 (mV)
VDD_MCU   : 749.0 (mV)
VDDIO_PVT_MCU : 1794.0 (mV)
VDDIO_TOP4_1V8 : 1794.0 (mV)
VDDQ_DDR0 : 504.0 (mV)
VDD2H_DDR0 : 1042.0 (mV)
VAA_DDR0 : 1794.0 (mV)
VDD_DDR0 : 743.0 (mV)
VDDQ_DDR1 : 504.0 (mV)
VDD2H_DDR1 : 1042.0 (mV)
VAA_DDR1 : 1794.0 (mV)
VDD_DDR1 : 743.0 (mV)
VDDIO_SD_SDIO_T : 1788.0 (mV)
VDD_CPU   : 776.0 (mV)
VAA_DDR2 : 1780.0 (mV)
VDD2H_DDR2 : 1030.0 (mV)
VDD_DDR2 : 737.0 (mV)
VDDIO_TOP2_1V8 : 1780.0 (mV)
VDD_BPU   : 737.0 (mV)
VDDIO_TOP0_1V8 : 1780.0 (mV)
PLL_TOP0_VDDHV : 1768.0 (mV)
VDDIO_SD_33_MCU : 3304.0 (mV)
VDDIO_ADC_MCU : 1800.0 (mV)
VDDIO_MCU_1V8 : 1800.0 (mV)
PLL_MCU_VDDHV : 1800.0 (mV)
PLL_MCU_VDDPOST : 746.0 (mV)
PLL_MCU_VDDREF : 751.0 (mV)
VDD_TOP    : 757.0 (mV)
VDD_AON    : 751.0 (mV)
VDDIO_SD_AON_CA : 1662.0 (mV)
VDDIO_PVT_1V8 : 1800.0 (mV)
cpu frequency-->
           min   cur   max
policy0: 1125000 1500000 1500000
policy4: 1125000 1500000 1500000
bpu status information---->
           ratio
bpu0:     0

```

temperature:

- **pvt_cmn_pvtc1_t1/2**: CPU temperature, in degrees Celsius (°C).
- **pvt_mcu_pvtc1_t1/2**: MCU temperature, in degrees Celsius (°C).
- **pvt_bpu_pvtc1_t1**: BPU temperature, in degrees Celsius (°C).

cpu frequency:

- **min**: Minimum operating frequency of the CPU.
- **cur**: Current operating frequency of the CPU.
- **max**: Maximum operating frequency of the CPU.
- This information shows the frequency range (minimum, current, and maximum) for each CPU cluster.

bpu status information:

- `ratio`: BPU load ratio during operation.
- This information indicates the BPU load.

4. rdkos_info

The **rdkos_info command** is used to collect, in one go, software and hardware versions of the `RDK` system, a list of loaded drivers, installed RDK software packages, and the latest system logs, enabling users to quickly obtain the current system status information.

• Syntax

```
sudo rdkos_info [options]
```

• Options

All options are optional. If run without any options, `rdkos_info` defaults to the concise output mode.

- `-b`: Basic output mode; system logs will not be collected.
- `-s`: Concise output mode (default); outputs the latest 30 lines of system logs.
- `-d`: Detailed output mode; outputs the latest 300 lines of system logs.
- `-v`: Display version information.
- `-h`: Display help information.

• Common Usage

Default usage:

```
sudo rdkos_info
```

Partial output is shown below:

```
sunrise@ubuntu:/root$ sudo rdkos_info
===== RDK System Information Collection =====

[Hardware Model]:
    D-Robotics RDK S600 V0P5 (Board Id = 0x6A84)

[CPU And BPU Status]:
=====1=====
temperature-->
    pvt_cmn_pvtc1_t1 : 38.932 (C)
    pvt_cmn_pvtc1_t2 : 40.587 (C)
    pvt_mcu_pvtc1_t1 : 38.013 (C)
    pvt_mcu_pvtc1_t2 : 38.381 (C)
    pvt_bpu_pvtc1_t1 : 39.300 (C)
voltage-->
    FAKE      : 100.0 (mV)
    VDDQ_DDR2 : 504.0 (mV)
    VDD_MCU   : 749.0 (mV)
    VDDIO_PVT_MCU : 1794.0 (mV)
    VDDIO_TOP4_1V8 : 1794.0 (mV)
    VDDQ_DDR0 : 498.0 (mV)
    VDD2H_DDR0 : 1042.0 (mV)
    VAA_DDR0 : 1794.0 (mV)
```

```

VDD_DDR0 : 743.0 (mV)
VDDQ_DDR1 : 498.0 (mV)
VDD2H_DDR1 : 1052.0 (mV)
VAA_DDR1 : 1794.0 (mV)
VDD_DDR1 : 743.0 (mV)
VDDIO_SD_SDIO_T : 1788.0 (mV)
VDD_CPU : 776.0 (mV)
VAA_DDR2 : 1780.0 (mV)
VDD2H_DDR2 : 1042.0 (mV)
VDD_DDR2 : 737.0 (mV)
VDDIO_TOP2_1V8 : 1780.0 (mV)
VDD_BPU : 737.0 (mV)
VDDIO_TOP0_1V8 : 1780.0 (mV)
PLL_TOP0_VDDHV : 1768.0 (mV)
VDDIO_SD_33_MCU : 3304.0 (mV)
VDDIO_ADC_MCU : 1800.0 (mV)
VDDIO_MCU_1V8 : 1800.0 (mV)
PLL_MCU_VDDHV : 1800.0 (mV)
PLL_MCU_VDDPOST : 751.0 (mV)
PLL_MCU_VDDREF : 751.0 (mV)
VDD_TOP : 757.0 (mV)
VDD_AON : 751.0 (mV)
VDDIO_SD_AON_CA : 1662.0 (mV)
VDDIO_PVT_1V8 : 1800.0 (mV)
cpu frequency-->
      min   cur   max
policy0: 1125000      1500000 1500000
policy4: 1125000      1500000 1500000
bpu status information---->
      ratio
bpu0:  0

```

```

[Total Memory]:      4.7Gi
[Used Memory]:       1.6Gi
[Free Memory]:       1.5Gi

```

```

[RDk OS Version]:
4.0.2-Beta

```

```

[RDk Kernel Version]:
Linux ubuntu 6.1.112-rt43-DR-4.0.2-2506091317-g08c4b8-g2e7d75 #80 SMP
PREEMPT_RT Mon Jun  9 13:18:49 CST 2025 aarch64 aarch64 aarch64 GNU/Linux

```

```

[RDk Miniboot Version]:
4.0.2-20250605231718

```

5.hrut_socuid

The **hrut_socuid** command prints the UID (Unique Identifier) of the current SoC chip.

• Syntax

```
hrut_socuid
```

• Common Usage

```
sunrise@ubuntu:~$ hirut_socuid  
060c0b0d3090694108255c4c00001079
```

6.devmem

`devmem` is a command in BusyBox. It can read from and write to hardware registers by using the `mmap` function to map device memory into user space via the `mmap` method of the `/dev/mem` driver, thereby enabling read/write operations on physical addresses.

Syntax

```
devmem ADDRESS [WIDTH [VALUE]]
```

Read/write from physical address

ADDRESS Address to act upon
WIDTH width (8/16/...)
VALUE Data to be written

- **ADDRESS:** The physical address on which to perform the operation. This is a required parameter specifying the address to read from or write to.
- **WIDTH:** An optional parameter indicating the data width. It can be specified as 8, 16, or 32, defining the bit-width of the data to read or write. If omitted, it defaults to 32 bits.
- **VALUE:** An optional parameter representing the data value to write. If `WIDTH` is provided, `VALUE` must match the specified width. If `VALUE` is omitted, the command performs a read operation.

Common Commands

- Reading registers

```
Read 32-bit: devmem 0xa600307c 32  
Read 16-bit: devmem 0xa600307c 16  
Read 8-bit: devmem 0xa600307c 8
```

- Writing registers

```
write 32-bit: devmem 0xa6003078 32 0x1000100  
write 16-bit: devmem 0xa6003078 16 0x1234  
write 8-bit: devmem 0xa6003078 8 0x12
```